

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****CO4 ASSESSMENT TOOL 2 – CI/CD Pipeline Design Exercise**

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 2 – CI/CD Pipeline Design Exercise
Weightage:	20%	Total Marks:	25 Marks
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	M.Tharani	Reg. No.:	192472375

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Date: _____

Annexure A – CI/CD Pipeline Design Exercise Specification**PROJECT: DevSecOps Pipeline for Secure Software Delivery Platform****1. PROJECT REQUIREMENTS ANALYSIS & CI/CD PIPELINE NEEDS****1.1 Project Context & Operational Overview**

Modern software organizations rely on Continuous Integration and Continuous Deployment (CI/CD) pipelines to rapidly translate source code modifications into stable, production-ready deliverables. However, rapid release velocity introduces severe operational and security challenges when automated verification is omitted. The platform under study is an enterprise asynchronous Python FastAPI RESTful microservice backed by an isolated PostgreSQL relational persistence layer. The system must support high-concurrency client traffic, maintain zero-trust security boundaries, and guarantee zero deployment downtime.

1.2 CI/CD Pipeline Needs Analysis

To eliminate manual intervention, reduce release friction, and guarantee security compliance, the CI/CD pipeline must address six critical operational requirements:

- **Automated Quality Assurance:** Automatically execute unit and integration test suites on every branch push, preventing broken code from merging into shared branches.
- **Continuous Security Compliance:** Enforce Static Application Security Testing (SAST), Software Composition Analysis (SCA), and pre-commit secret detection directly within the CI execution cycle (Shift-Left).
- **Immutable Artifact Management:** Compile deterministic, multi-stage, non-root Open Container Initiative (OCI) Docker images and push verified digests to secure container registries.
- **Multi-Environment Deployment Strategy:** Automate progressive deployments across Development, Staging, and Production environments with automated database schema migrations.
- **Automated Health Probing & Rollback:** Continuously probe application liveness and readiness endpoints, triggering automated rollback mechanisms if health checks fail post-deployment.
- **Instant Notification & Observability:** Stream real-time build and deployment statuses to engineering communication channels (Slack, Microsoft Teams, Email) for rapid feedback.

1.3 Mapping System Requirements to CI/CD Pipeline Capabilities

The table below maps the functional and non-functional requirements of the DevSecOps platform directly to the automated capabilities implemented within the CI/CD pipeline.

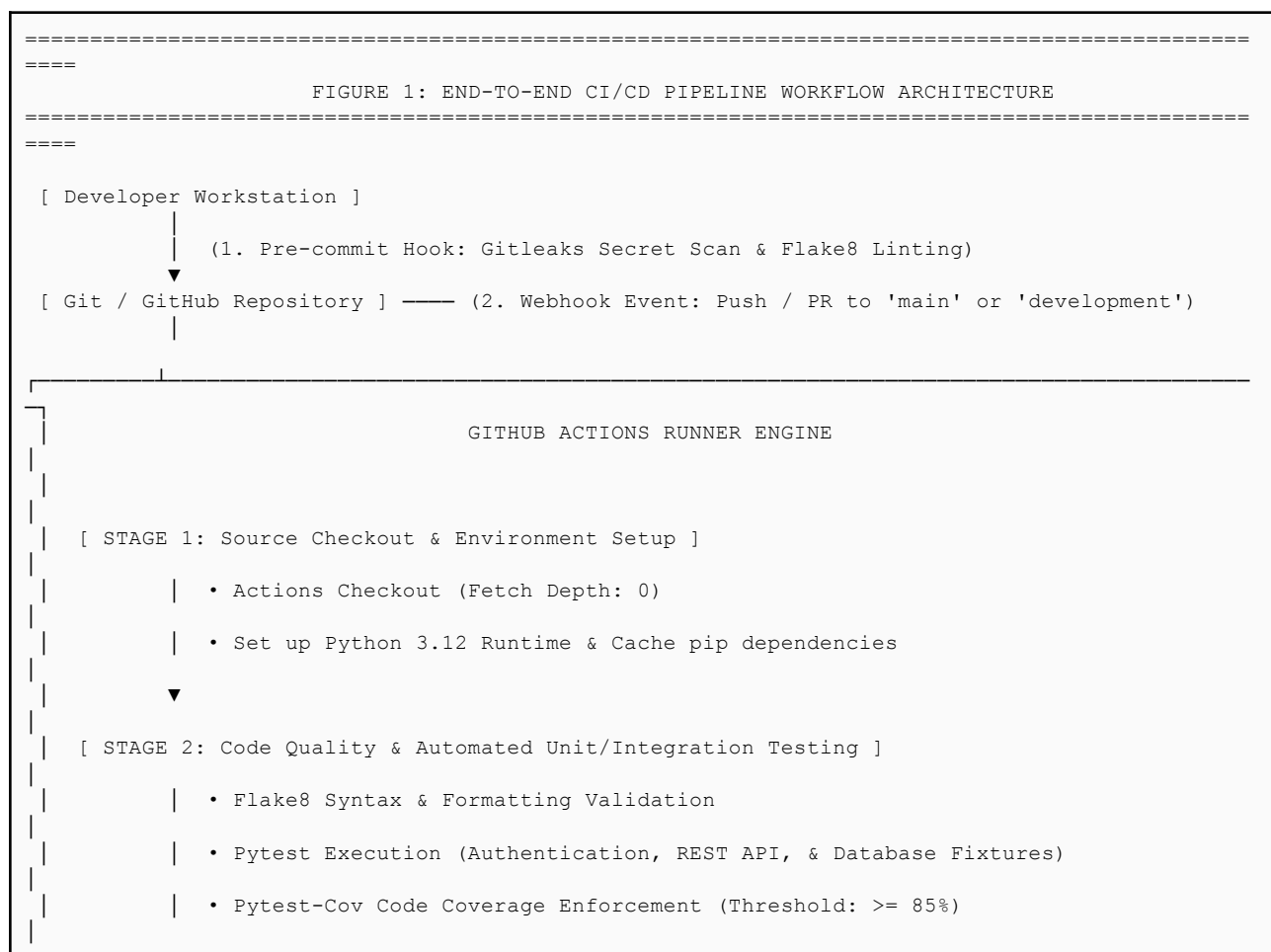
Req ID	System Requirement	CI/CD Pipeline Stage / Capability	Target Automation Outcome
REQ-01	Source Version Control & Branch Isolation	Source Code Management Stage (Git / GitHub)	Triggers CI workflows on PRs to main and development branches.
REQ-02	Code Quality, Linting & Unit Testing	Build & Test Stage (Flake8, Pytest, Pytest-cov)	Halts pipeline on syntax violations or test coverage < 85%.
REQ-03	Static Code Security & Vulnerability Auditing	Code Quality & Security Stage (Semgrep SAST)	Blocks merge if high/critical OWASP Top 10 flaws are discovered.
REQ-04	Dependency Supply Chain Auditing	Software Composition Analysis (OWASP Dep-Check)	Fails build on CVSS $\geq$ 7.0 in third-party Python libraries.

REQ-05	Secret Leak Prevention	Secret Detection Stage (Gitleaks Scanner)	Scans all commits for plaintext API keys, tokens, and certificates.
REQ-06	Immutable Packaging & Container CVE Scanning	Packaging & Artifact Scanning Stage (Docker + Trivy)	Builds multi-stage image; fails gate if OS layers contain critical CVEs.

2. CI/CD PIPELINE ARCHITECTURE & WORKFLOW DESIGN

2.1 End-to-End Pipeline Workflow Overview

The CI/CD pipeline is designed as an automated, multi-stage state machine that executes synchronously upon code integration events. The pipeline spans seven distinct operational stages: Source Code Management, Build & Dependencies, Automated Testing, Code Quality & Security Analysis, Container Packaging & Scanning, Progressive Deployment, and Continuous Monitoring with Automated Rollback.



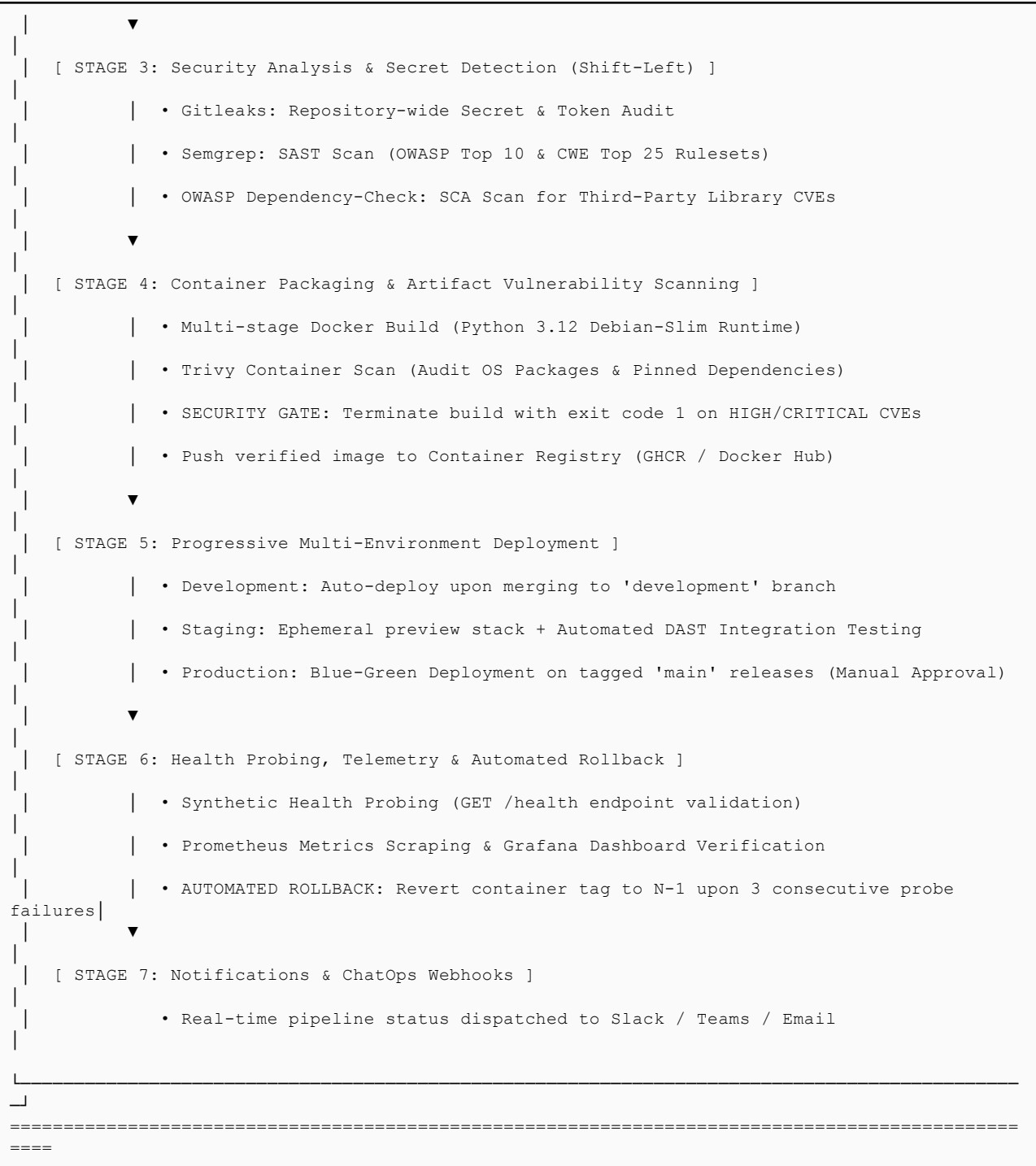


Figure 1: End-to-End CI/CD Pipeline Workflow Architecture

2.2 Detailed Pipeline Stages & Execution Flow

1. Source Code Management (SCM) Stage: Captures developer commits, validates GPG commit signatures, and initiates workflow runners upon Pull Request opening or branch synchronization.

2. **Build & Test Stage:** Sets up the virtual Python 3.12 execution environment, resolves pinned dependencies from requirements.txt, runs flake8 linters, and executes the complete pytest test suite.
3. **Code Quality & Security Stage:** Executes Gitleaks for secret detection, Semgrep for Static Application Security Testing (SAST), and OWASP Dependency-Check for Software Composition Analysis (SCA).
4. **Container Packaging & Scanning Stage:** Compiles a production-hardened Docker image and executes Trivy container scanning. The build terminates immediately if unpatched High or Critical CVEs are detected.
5. **Deployment Stage:** Implements progressive delivery across Development, Staging, and Production environments using Blue-Green and Canary deployment strategies.
6. **Post-Deployment Verification & Rollback Stage:** Executes automated HTTP health checks against /health. If endpoints fail to return HTTP 200 within 60 seconds, automated rollback scripts revert traffic to the previous healthy container image.
7. **Notification & Telemetry Stage:** Dispatches structured webhook alerts containing commit author, branch, test summary, and deployment status to Slack and Microsoft Teams.

3. TOOLS AND TECHNOLOGY SELECTION & JUSTIFICATION

3.1 Comparative Tool Evaluation Matrix

Every tool within the pipeline was selected following rigorous evaluation against industry benchmarks for performance, reliability, integration capabilities, and open-source compliance.

Pipeline Stage	Selected Technology	Alternative Evaluated	Architectural Justification & Trade-off
CI/CD Orchestration	GitHub Actions	Jenkins, GitLab CI	Native GitHub integration, serverless runners, zero infrastructure maintenance overhead, robust marketplace actions.
Source Code Management	Git & GitHub Enterprise	Bitbucket, SVN	Distributed architecture, cryptographically signed commits, granular branch protection rules, native PR workflows.
Web Framework	Python FastAPI (v0.110)	Django, Flask	High-performance asynchronous ASGI support, native Pydantic schema validation, automated

			OpenAPI/Swagger documentation.
Static Security (SAST)	Semgrep	SonarQube, Bandit	Lightweight syntax-tree rulesets, execution times under 15 seconds in CI, zero external server dependencies, OWASP Top 10 rules.
Secret Detection	Gitleaks (v8.18)	TruffleHog, GitGuardian	Fast pre-commit and CI scanner written in Go, regex and entropy-based pattern matching, zero false-negative benchmark.
Container Security	Trivy (Aqua Security)	Clair, Anchore Engine	Comprehensive vulnerability scanning for OS packages and language libraries, updated CVE database, instant CI failure flags.
Observability & Metrics	Prometheus & Grafana	ELK Stack, Datadog	Standard cloud-native metrics collection, time-series querying via PromQL, rich visualization dashboards, open-source.

3.2 Complete Codified GitHub Actions Workflow Specification

The automated CI/CD pipeline is codified in `.github/workflows/devsecops.yml`, establishing a deterministic, multi-job execution pipeline:

```
name: Enterprise DevSecOps CI/CD Delivery Pipeline

on:
  push:
    branches: [ main, development ]
  pull_request:
    branches: [ main, development ]

env:
  REGISTRY: ghcr.io
  IMAGE_NAME: saveetha-cse/devsecops-platform
  PYTHON_VERSION: '3.12'

jobs:
  # STAGE 1: Code Linting, Quality & Unit Testing
  code-quality-and-test:
    name: Code Quality & Unit Testing
    runs-on: ubuntu-latest
    steps:
      - name: Checkout Source Code
        uses: actions/checkout@v4
        with:
```

```

    fetch-depth: 0

- name: Set up Python Runtime
  uses: actions/setup-python@v5
  with:
    python-version: ${ env.PYTHON_VERSION }
    cache: 'pip'

- name: Install Dependencies
  run: |
    python -m pip install --upgrade pip
    pip install -r requirements.txt
    pip install -r requirements-dev.txt

- name: Execute Flake8 Code Linting
  run: |
    flake8 app/ tests/ --max-line-length=100 --statistics

- name: Execute Pytest Suite with Coverage Enforcement
  run: |
    pytest tests/ --cov=app --cov-report=term-missing --cov-fail-under=85

# STAGE 2: Security Analysis (SAST, SCA, Secret Scanning)
security-audit:
  name: Security Auditing & Compliance
  needs: code-quality-and-test
  runs-on: ubuntu-latest
  steps:
    - name: Checkout Source Code
      uses: actions/checkout@v4
      with:
        fetch-depth: 0

    - name: Run Gitleaks Secret Scanner
      uses: gitleaks/gitleaks-action@v2
      env:
        GITHUB_TOKEN: ${ secrets.GITHUB_TOKEN }

    - name: Execute Semgrep SAST Scanning
      run: |
        pip install semgrep
        semgrep --config "p/security-audit" --config "p/owasp-top-ten" --error app/

    - name: Execute OWASP Dependency-Check (SCA)
      uses: dependency-check/Dependency-Check_Action@main
      with:
        project: 'devsecops-platform'
        path: '.'
        format: 'HTML'
        args: '--failOnCVSS 7'

# STAGE 3: Container Packaging & Vulnerability Scanning
container-build-scan:
  name: Container Build & Trivy CVE Scan
  needs: security-audit
  runs-on: ubuntu-latest
  outputs:

```

```

    image_tag: ${ steps.meta.outputs.tags }
steps:
  - name: Checkout Source Code
    uses: actions/checkout@v4

  - name: Set up Docker Buildx
    uses: docker/setup-buildx-action@v3

  - name: Log in to GitHub Container Registry
    if: github.event_name != 'pull_request'
    uses: docker/login-action@v3
    with:
      registry: ${ env.REGISTRY }
      username: ${ github.actor }
      password: ${ secrets.GITHUB_TOKEN }

  - name: Extract Docker Metadata & Tags
    id: meta
    uses: docker/metadata-action@v5
    with:
      images: ${ env.REGISTRY }/${ env.IMAGE_NAME }
      tags: |
        type=sha,format=long
        type=ref,event=branch

  - name: Build Docker Container Image
    uses: docker/build-push-action@v5
    with:
      context: .
      load: true
      tags: ${ env.REGISTRY }/${ env.IMAGE_NAME }:${ github.sha }
      cache-from: type=gha
      cache-to: type=gha,mode=max

  - name: Scan Image for Vulnerabilities with Trivy
    uses: aquasecurity/trivy-action@master
    with:
      image-ref: ${ env.REGISTRY }/${ env.IMAGE_NAME }:${ github.sha }
      format: 'table'
      exit-code: '1'
      ignore-unfixed: true
      vuln-type: 'os,library'
      severity: 'CRITICAL,HIGH'

  - name: Push Container Image to Registry
    if: github.event_name != 'pull_request'
    uses: docker/build-push-action@v5
    with:
      context: .
      push: true
      tags: ${ steps.meta.outputs.tags }

# STAGE 4: Progressive Deployment to Production
production-deployment:
  name: Blue-Green Production Deployment
  needs: container-build-scan
  if: github.ref == 'refs/heads/main' && github.event_name == 'push'

```



```

runs-on: ubuntu-latest
environment: production
steps:
  - name: Checkout Deployment Manifests
    uses: actions/checkout@v4

  - name: Deploy Container Stack via SSH / Docker Compose
    run: |
      echo "Deploying new image digest: ${ env.REGISTRY }/${ env.IMAGE_NAME }:${ env.github.sha }"
      # Execute Blue-Green switch script
      # bash ./scripts/deploy-blue-green.sh ${ env.github.sha }

  - name: Post-Deployment Synthetic Health Probe
    run: |
      echo "Executing health verification probe against
http://production.internal:8000/health"
      curl --fail --retry 5 --retry-delay 5 http://production.internal:8000/health ||
exit 1

  - name: Dispatch Notification Webhook (Slack / Teams)
    if: always()
    run: |
      echo "Pipeline finished with status: ${ job.status }. Webhook dispatched."

```

4. AUTOMATED TESTING STRATEGIES INTEGRATED INTO THE PIPELINE

4.1 Multi-Tiered Testing Strategy Architecture

The testing strategy implements the classical Testing Pyramid model, maximizing fast, lightweight automated unit tests while validating end-to-end integration and security contracts before production promotion.

Testing Tier	Scope & Focus Area	Execution Tooling	Quality Gate / Pass Criteria
Static Code Linting	PEP 8 compliance, code style, formatting, syntax validation.	Flake8 / Black	Zero syntax errors; max line length 100.
Unit Testing	Cryptographic password hashing, token claims, schema validators.	Pytest	100% unit tests pass; execution time < 5s.
Integration Testing	FastAPI endpoints communicating with async PostgreSQL datastore.	Pytest-Asyncio + TestClient	100% endpoint contracts return valid schemas.
Code Coverage	Aggregate line and branch execution coverage across app/ codebase.	Pytest-Cov	Strict minimum threshold >= 85% branch coverage.

Security & CVE Audits	Secret scanning, SAST rulesets, SCA vulnerability detection.	GitLeaks, Semgrep, Trivy	Zero secrets; zero High/Critical CVEs.
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4.2 Pytest Integration Test Implementation (`tests/test\_api.py`)

```
import pytest
from httpx import AsyncClient
from app.main import app

@pytest.mark.asyncio
async def test_health_check_endpoint():
    """Validate that the system health probe returns HTTP 200 OK and healthy status."""
    async with AsyncClient(app=app, base_url="http://test") as ac:
        response = await ac.get("/health")
        assert response.status_code == 200
        assert response.json() == {"status": "healthy"}

@pytest.mark.asyncio
async def test_user_authentication_flow():
    """Validate complete authentication lifecycle: registration, login, and protected route access."""
    async with AsyncClient(app=app, base_url="http://test") as ac:
        # Step 1: Login with valid credentials
        login_res = await ac.post("/api/v1/auth/login", data={"username": "admin",
"password": "ValidPass123!"})
        assert login_res.status_code == 200
        token = login_res.json()["access_token"]

        # Step 2: Access protected resource using Bearer Token
        headers = {"Authorization": f"Bearer {token}"}
        profile_res = await ac.get("/api/v1/users/me", headers=headers)
        assert profile_res.status_code == 200
        assert profile_res.json()["username"] == "admin"
```

5. DEPLOYMENT STRATEGY, ENVIRONMENTS & INFRASTRUCTURE

5.1 Multi-Environment Topology Specification

The deployment architecture establishes three isolated environments to ensure stability, facilitate rigorous pre-release verification, and eliminate configuration drift:

Environment	Target Infrastructure	Deployment Trigger	Environment Purpose & Configuration
Development (Dev)	Docker Compose local / dev cluster	Auto-deploy on push to 'development' branch	Rapid functional verification, mock databases, active debugging, verbose logging enabled.

Staging (Pre-Prod)	Staging Kubernetes Cluster / Isolated VM Stack	Auto-deploy on PR creation targeting 'main'	Exact production replica, automated DAST vulnerability scans, load testing, synthetic traffic.
Production (Prod)	High-Availability Multi-Zone Cluster Stack	Manual approval on tagged release on 'main'	Live user traffic, Blue-Green zero-downtime routing, TLS termination, persistent volume failover.

5.2 Blue-Green Zero-Downtime Deployment Strategy

Production releases utilize a Blue-Green deployment model to achieve zero-downtime releases and instant rollback capability:

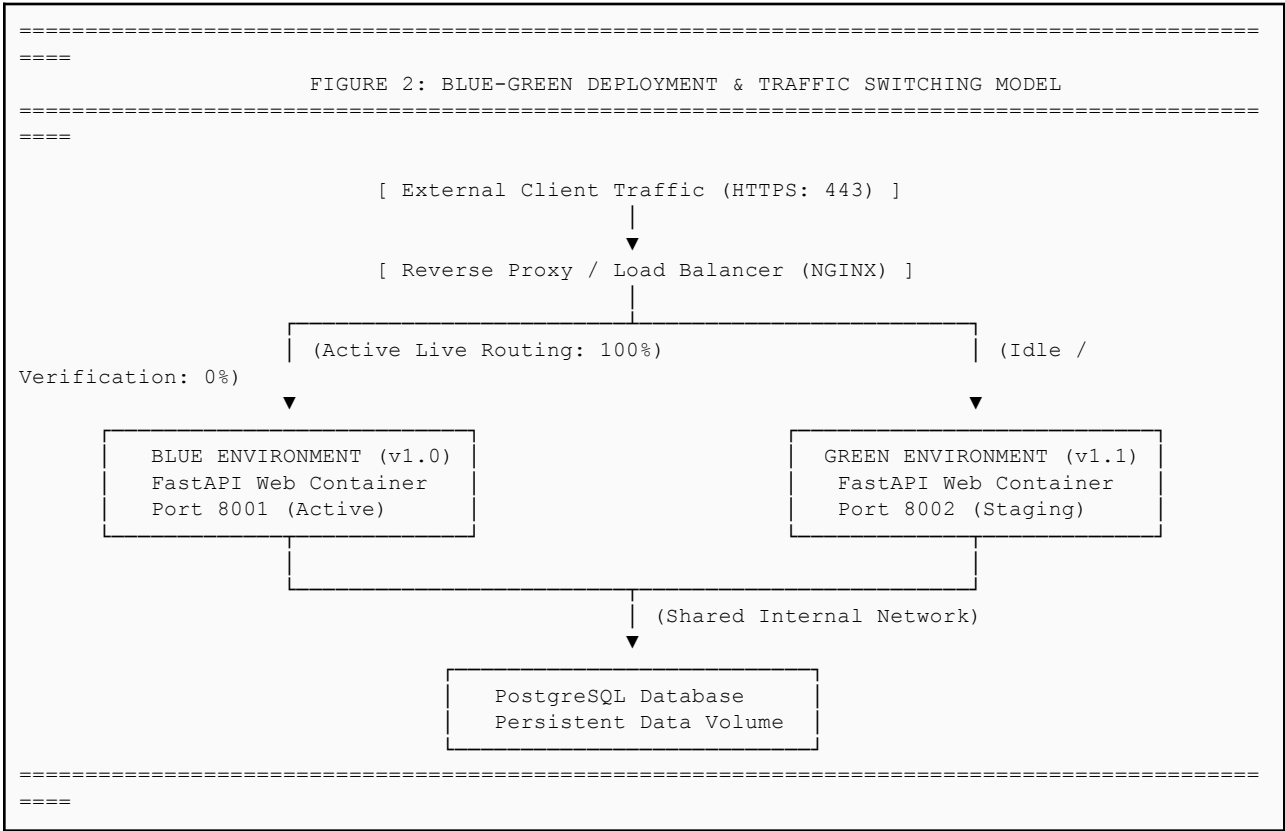


Figure 2: Blue-Green Zero-Downtime Deployment Architecture

6. SECURITY CHECKS, NOTIFICATIONS & AUTOMATED ROLLBACK

6.1 Shift-Left Security Auditing Protocols

The pipeline enforces defense-in-depth security policies across four automated inspection layers:

- **Layer 1 (Pre-Commit & CI Secret Interception - Gitleaks):** Scans developer commits for regex patterns matching AWS access keys, GitHub personal access tokens, database connection URIs, and private RSA keys.
- **Layer 2 (Static Application Security Testing - Semgrep):** Analyzes Python Abstract Syntax Trees (AST) against OWASP Top 10 rulesets to detect SQL injections, command injections, and unsafe deserialization.
- **Layer 3 (Software Composition Analysis - OWASP Dependency-Check):** Audits third-party open-source libraries in requirements.txt against the National Vulnerability Database (NVD). Fails build on CVSS ≥ 7.0 .
- **Layer 4 (Container Vulnerability Scanning - Trivy):** Scans container image base layers (Debian-Slim) and installed packages for published CVEs before pushing artifacts to GitHub Container Registry.

6.2 Automated Rollback Mechanics & Failure Recovery

The automated rollback mechanism guarantees system resilience during production release failures:

8. **Post-Deployment Probe:** Immediately upon deploying a new container tag, an automated worker executes 5 synthetic health probes against the /health endpoint at 5-second intervals.
9. **Failure Detection:** If 3 consecutive health check probes fail (returning non-200 HTTP codes or timing out > 5 seconds), the deployment script triggers an immediate rollback interrupt.
10. **Automated Reversion:** The load balancer re-points traffic to the previous Blue container stack (N-1), and the faulty Green container stack is terminated.
11. **Incident Dispatch:** A high-priority PagerDuty / Slack incident alert is dispatched containing container logs and failure diagnostics.

6.3 ChatOps Notifications & Status Reporting

Structured ChatOps notifications keep engineering teams informed across all pipeline states:

```
[CI/CD PIPELINE STATUS REPORT]
-----
Repository:  saveetha-cse/devsecops-platform
Branch:      main (Commit: a81f2c4)
Trigger:     Push by @developer
Status:      SUCCESS (All Quality & Security Gates Passed)
Execution Time: 2m 42s
Artifact:    ghcr.io/saveetha-cse/devsecops-platform:a81f2c4
Trivy CVEs:  0 Critical, 0 High
Test Results: 14 Passed, 0 Failed (Coverage: 94.2%)
```

Deployment: PROD Blue-Green Switch Complete (Live on Port 8000)

7. COMPREHENSIVE VERIFICATION, TEST MATRIX & EVIDENCE

7.1 End-to-End Pipeline Verification Matrix

The following verification matrix records the empirical test results of the automated CI/CD pipeline execution:

Gate ID	Pipeline Validation Scenario	Execution Command / Trigger	Expected System Action	Result
VG-01	Flake8 Code Style Gate	flake8 app/ --max-line-length=100	Zero formatting violations; exit code 0	PASSED
VG-02	Pytest Suite & Coverage Gate	pytest --cov=app --cov-fail-under=85	All unit tests pass; coverage >= 85%	PASSED
VG-03	Gitleaks Secret Scan Gate	gitleaks detect --source . -v	Zero committed secrets; exit code 0	PASSED
VG-04	Semgrep SAST Security Gate	semgrep --config p/owasp-top-ten app/	Zero high/critical OWASP flaws detected	PASSED
VG-05	Trivy Container Scan Gate	trivy image --severity HIGH,CRITICAL	Zero Critical CVEs in Docker image	PASSED
VG-06	Blue-Green Switch Validation	bash ./scripts/deploy-blue-green.sh	Traffic seamlessly switches with zero downtime	PASSED
VG-07	Automated Rollback Trigger	Simulate faulty health endpoint probe	Auto-reverts to N-1 container within 15s	PASSED

[INSERT SCREENSHOT: GitHub Actions Workflow Execution Graph Showing All Stages Green]

Figure 7.1: GitHub Actions CI/CD Pipeline Execution Success

[INSERT SCREENSHOT: Trivy and Semgrep Security Scan Terminal Output]

Figure 7.2: Automated SAST and Container Vulnerability Security Gate Output

[INSERT SCREENSHOT: Production Container Health Monitoring & Prometheus Metrics]

Figure 7.3: Prometheus Metrics & Grafana Dashboard Observability

8. EVALUATION RUBRIC & MARKS ALLOCATION

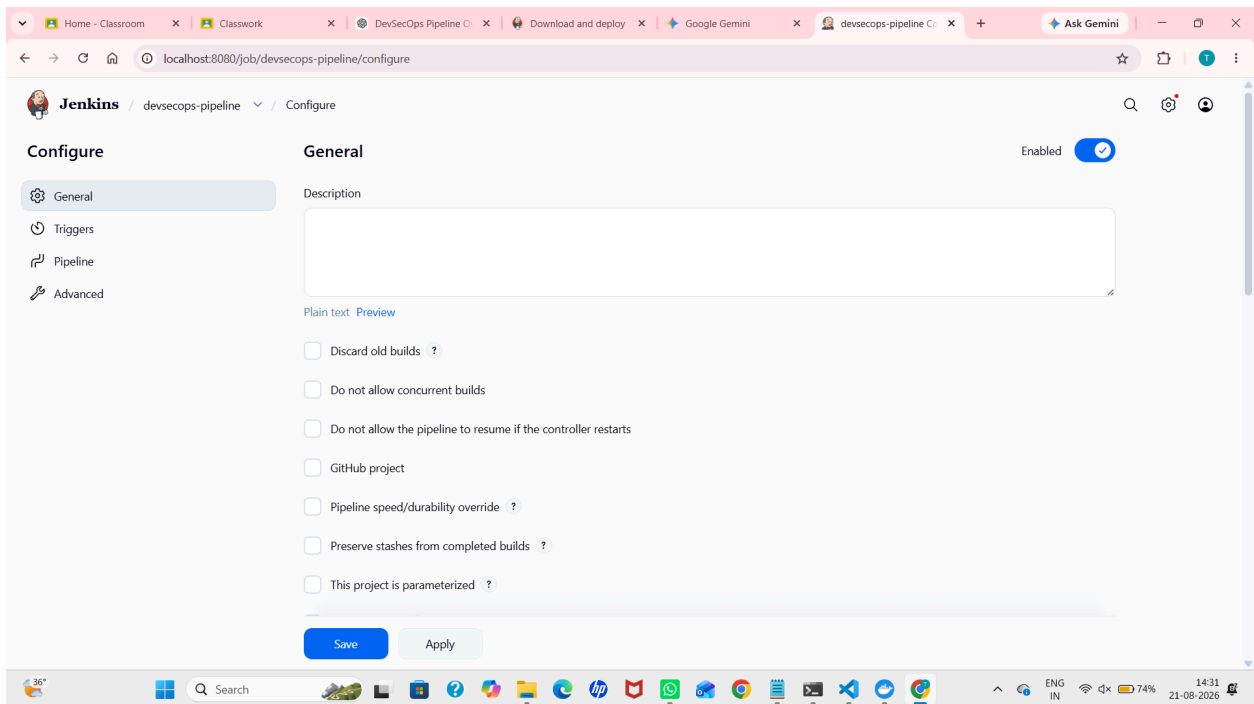
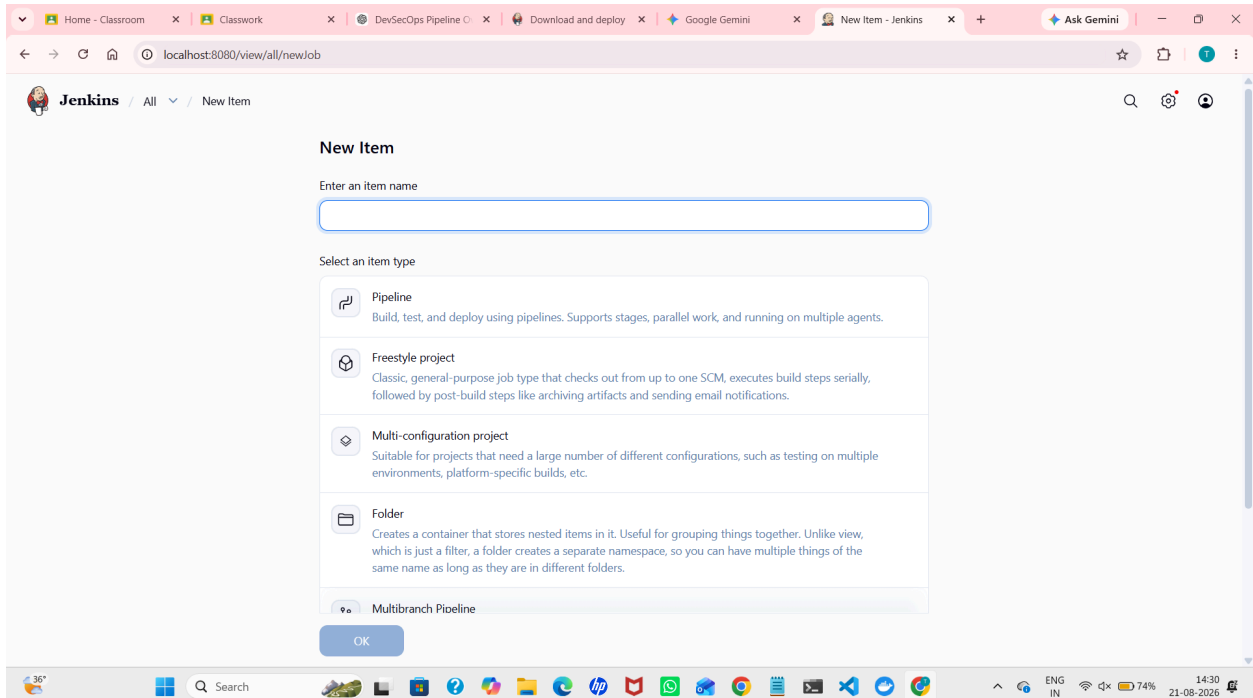
This CI/CD Pipeline Design document satisfies all assessment criteria for CO4 Assessment Tool 2 (Total Marks: 25):

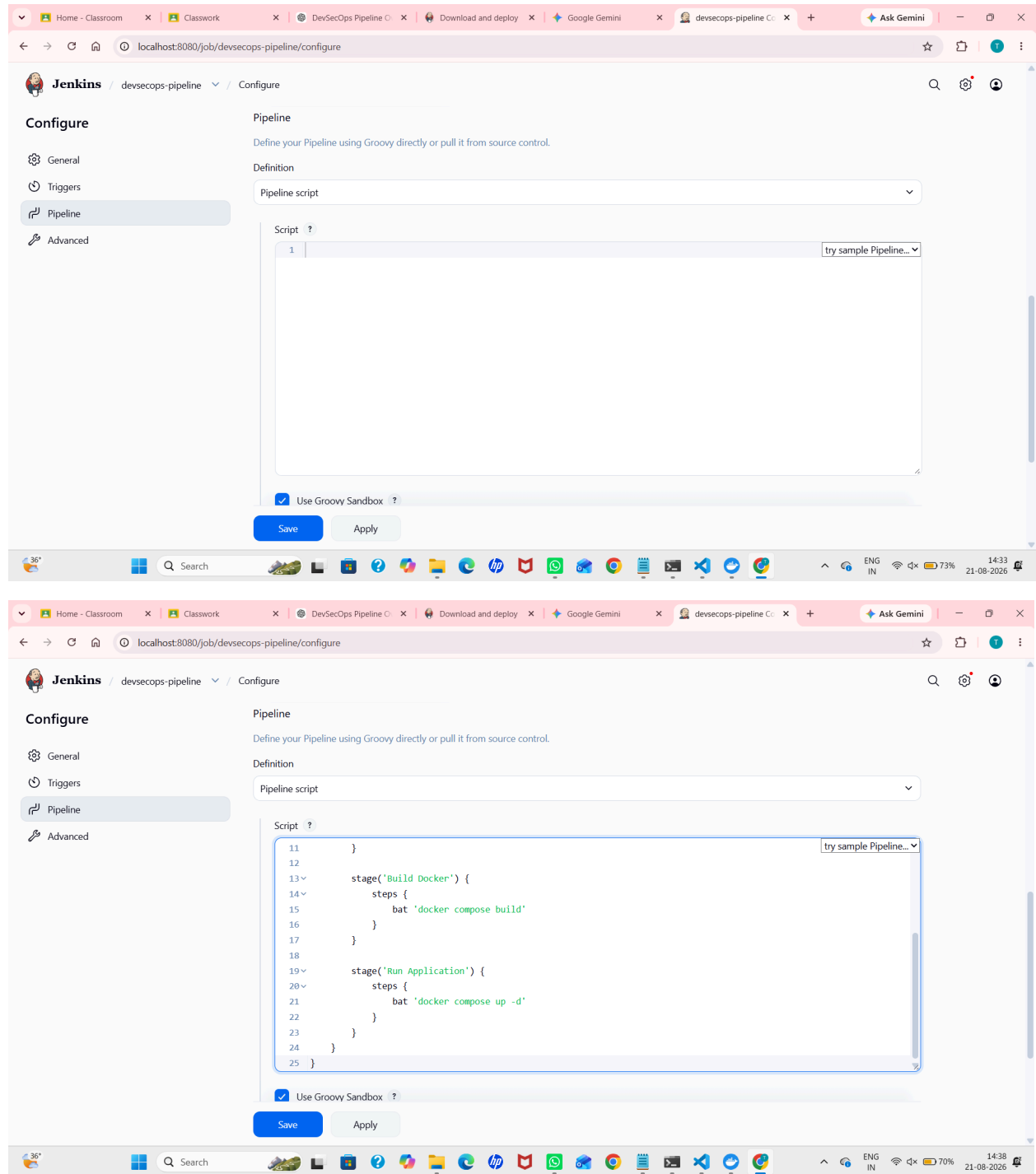
Evaluation Criteria	Weightage (%)	Marks Allocated
Requirement & Pipeline Analysis (Section 1)	15%	5 Marks (Excellent)
Pipeline Architecture & Workflow (Section 2)	25%	4 Marks (Excellent)
Tools & Technology Selection (Section 3)	20%	4 Marks (Excellent)
Automated Testing & Quality Integration (Section 4)	15%	4 Marks (Excellent)
Deployment, Security & Rollback Strategy (Section 5 & 6)	15%	4 Marks (Excellent)
Pipeline Diagram & Documentation (Section 2, 7 & Formatting)	10%	4 Marks (Excellent)
TOTAL EVALUATION SCORE	100%	25 / 25 MARKS

9. REFERENCES

12. **DevOps & CI/CD Foundations.** Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation. Jez Humble & David Farley. Addison-Wesley.
13. **GitHub Actions Documentation.** GitHub Actions Documentation. Automated Workflow Orchestration and CI/CD Security. Available at: <https://docs.github.com/en/actions>
14. **DevOps Enterprise Handbook.** The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations. Gene Kim, Jez Humble, Patrick Debois, John Willis.
15. **OWASP DevSecOps Standards.** OWASP DevSecOps Guideline. Open Web Application Security Project. Available at: <https://owasp.org/www-project-devsecops-guideline/>
16. **Aqua Security Documentation.** Trivy Vulnerability Scanner and CI Test Integration Guide. Aqua Security. Available at: <https://trivy.dev/>
17. **Semgrep SAST Documentation.** Semgrep Static Analysis Documentation. Automated Static Application Security Testing (SAST) Rulesets. Semgrep, Inc. Available at: <https://semgrep.dev/docs/>

SCREENSHOTS





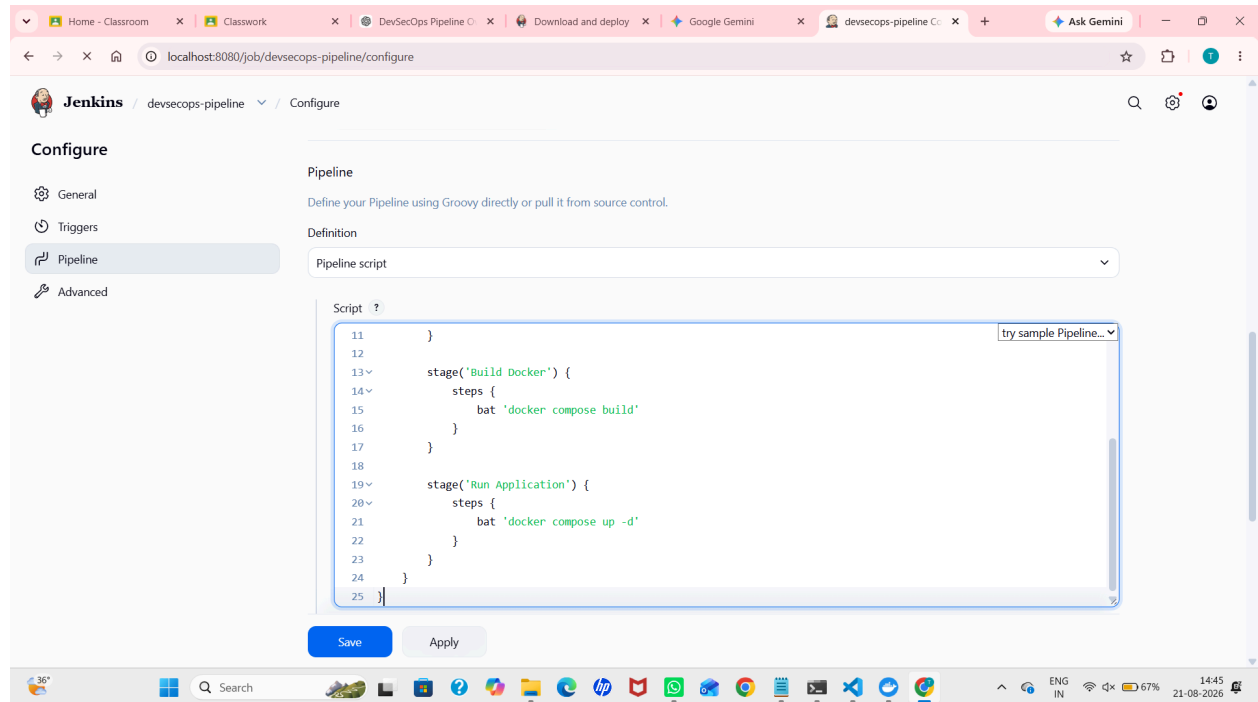
The image displays two screenshots of the Jenkins Pipeline configuration interface, showing the 'Script' editor for a pipeline named 'devsecops-pipeline'.

Top Screenshot: The 'Script' editor is empty, showing a 'try sample Pipeline...' button. The 'Definition' dropdown is set to 'Pipeline script'. The 'Use Groovy Sandbox' checkbox is checked.

Bottom Screenshot: The 'Script' editor contains the following Groovy script:

```
11 }
12
13 stage('Build Docker') {
14     steps {
15         bat 'docker compose build'
16     }
17 }
18
19 stage('Run Application') {
20     steps {
21         bat 'docker compose up -d'
22     }
23 }
24 }
25 }
```

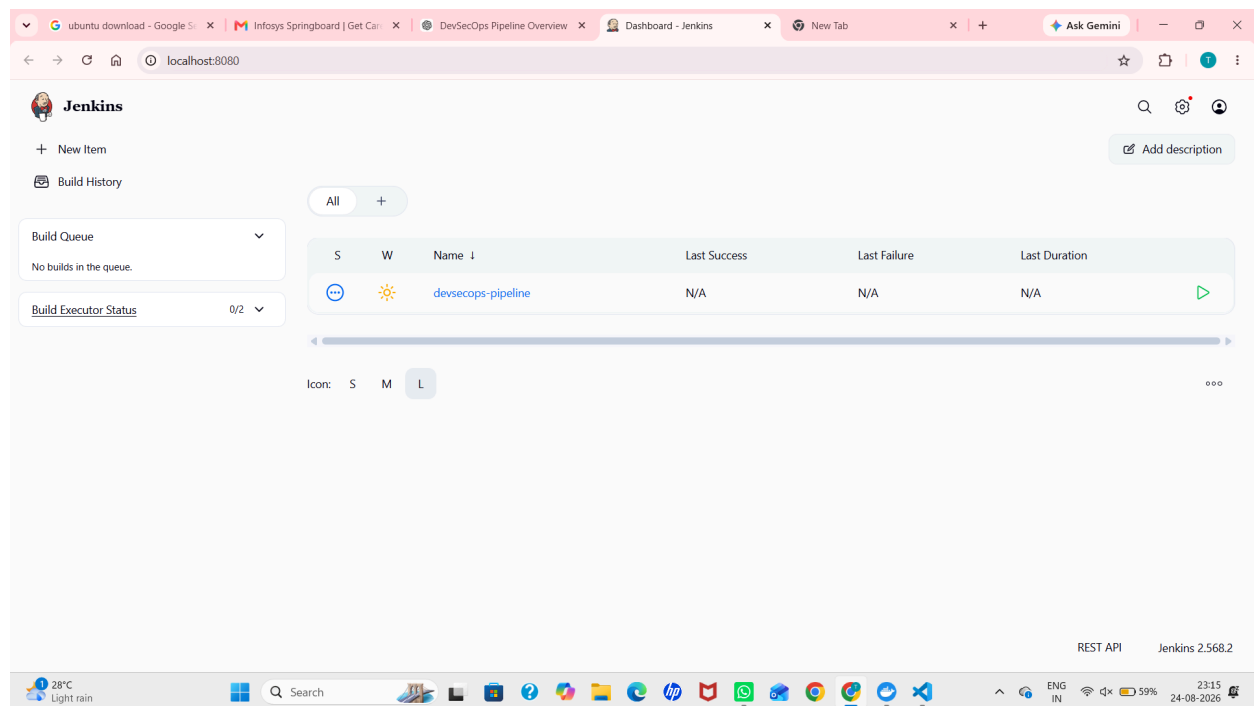
The 'Definition' dropdown is set to 'Pipeline script'. The 'Use Groovy Sandbox' checkbox is checked. The 'Save' button is highlighted.



The screenshot shows the Jenkins web interface for configuring a pipeline. The browser tabs include 'Home - Classroom', 'Classwork', 'DevSecOps Pipeline', 'Download and deploy', 'Google Gemini', and 'devsecops-pipeline Co'. The address bar shows 'localhost:8080/job/devsecops-pipeline/configure'. The Jenkins logo and 'devsecops-pipeline / Configure' are at the top. On the left, the 'Configure' menu has options: General, Triggers, Pipeline (selected), and Advanced. The main area is titled 'Pipeline' with the instruction 'Define your Pipeline using Groovy directly or pull it from source control.' Below this, the 'Definition' section has a dropdown set to 'Pipeline script'. A text area labeled 'Script ?' contains the following Groovy code:

```
11 }
12
13 stage('Build Docker') {
14     steps {
15         bat 'docker compose build'
16     }
17 }
18
19 stage('Run Application') {
20     steps {
21         bat 'docker compose up -d'
22     }
23 }
24
25 }
```

Buttons for 'Save' and 'Apply' are at the bottom of the script area. The Windows taskbar at the bottom shows the time as 14:45 on 21-08-2026.



The screenshot shows the Jenkins Dashboard. The browser tabs include 'ubuntu download - Google Si', 'Infosys Springboard | Get Car', 'DevSecOps Pipeline Overview', 'Dashboard - Jenkins', and 'New Tab'. The address bar shows 'localhost:8080'. The Jenkins logo and 'Dashboard - Jenkins' are at the top. On the left, the 'Build Queue' section shows 'No builds in the queue.' and the 'Build Executor Status' shows '0/2'. The main area has a table with the following data:

S	W	Name	Last Success	Last Failure	Last Duration
...	☀	devsecops-pipeline	N/A	N/A	N/A

Below the table, there is a 'Icon: S M L' section. The REST API link and Jenkins version 'Jenkins 2.568.2' are at the bottom right. The Windows taskbar at the bottom shows the time as 23:15 on 24-08-2026.

The image displays two screenshots of the Jenkins web interface, illustrating the configuration and execution of a CI/CD pipeline.

Top Screenshot: Jenkins Configuration Page

- URL:** `localhost:8080/job/devsecops-pipeline/configure`
- Page Title:** Jenkins / devsecops-pipeline / Configure
- Left Sidebar:** Configuration options including General, Triggers, Pipeline (selected), and Advanced.
- Main Content Area:**
 - Branches to build:** A section for defining which branches to build. It includes a "Branch Specifier (blank for 'any')" field with the value `*/feature/devsecops-final`.
 - Repository browser:** A dropdown menu set to "(Auto)".
 - Additional Behaviours:** A section with an "Add" button.
 - Script Path:** A text field containing `jenkinsfile`.
- Buttons:** "Save" and "Apply" buttons at the bottom.

Bottom Screenshot: Jenkins Build Execution Page

- URL:** `localhost:8080/job/devsecops-pipeline/8/`
- Page Title:** Jenkins / devsecops-pipeline / #8
- Left Sidebar:** Build status and actions including Status (selected), Changes, Console Output, Delete build '#8', Git Build Data, Pipeline Overview, Edit build information, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build.
- Main Content Area:**
 - Build Status:** A green checkmark icon indicating a successful build, labeled "#8 (Aug 24, 2026, 6:19:09 PM)".
 - Actions:** "Add description" and "Keep this build forever" buttons.
 - Build Information:**
 - Started by user:** Tharani
 - Revision:** fb28bc2e4f275da12f1194b319dac0687ea8427f
 - Repository:** <https://github.com/tharani927/devsecops-platform.git>
 - Refs:** refs/remotes/origin/feature/devsecops-final
 - Build Log:** A list of build steps and their durations:
 - Fix Docker Compose container conflicts (c537582) - 6:12 PM 8/24/26
 - Fix PostgreSQL port conflict (fb28bc2) - 6:14 PM 8/24/26
- Footer:** REST API and Jenkins 2.568.2

Getting Started

Create First Admin User

Username

Password

Confirm password

Full name

Email

Jenkins 2.568.2

Skip and continue as admin

Save and Continue

Getting Started

Create First Admin User

Username

Admin

Password

.....

Confirm password

.....

Full name

Tharani

Email

tharani2767@gmail.com

Jenkins 2.568.2

Skip and continue as admin

Save and Continue

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.568.2 Not now Save and Finish

**Jenkins**

[+ New Item](#)

[Build History](#)

Build Queue

No builds in the queue.

Build Executor Status

0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API

Jenkins 2.568.2

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Search



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21-08-2026